QUIZ 06 PRACTICE PROBLEMS – Solutions

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday, but are rather designed to be a starting point for your studies.

Some General Advice on Studying for this Quiz:

- Review Lecture Slides 13 and 14 carefully. There will be questions on interpreting odds and calculating odds ratios.
- Review Lecture Activities 13 and 14, and ensure you successfully achieved the stated learning objectives for each
- Check the solutions for Lecture Activities 13 and 14, along with Homework 7 (when they are released)

Use the following setup for Questions 1 - 8

Can regular exercise reduce the prevalence of hypertension (high blood pressure)? To answer this, experimenters recruited 100 US adult volunteers. 40 of the volunteers self-identified as leading "active" lifestyles, whereas the remaining 60 self-identified as leading "sedentary" lifestyles. The experimenters took readings of all 100 subject's blood pressures, and labeled each subject as either "hypertensive" or "not hypertensive."

Let's define the two variables as "**lifestyle**" (with levels "active" and "sedentary," measuring whether the subject was assigned to regularly exercise or not) and "**hypertension status**" (with levels "hypertensive" and "not hypertensive"). Additionally, whenever necessary, we'll define an "active lifestyle" as a "success" and "hypertensive" as a "success." Furthermore, we'll use "**control group**" to refer to the group of volunteers leading sedentary lifestyles, and will use the order Control — Treatment whenever necessary.

Shown below is a two-way table of the results of this study.

	Sedentary	Active	Total
Hypertensive	40	5	**60**
Not Hypertensive	20	35	**40**
Total	**60**	**40**	**100**

Table 1: Two-Way Table of Results

1. Which of the following best describes an observational unit in the context of this dataset?

(A) One US adult

(**B**) One US adult who leads an active lifestyle

(**C**) One US adult who leads a sedentary lifestyle

(**D**) One US adult who is hypertensive

(**E**) One US adult who is not hypertensive

2. What is the response variable in this study?

(**A**) Lifestyle **(B) Hypertension status**

Solution: We wish to determine whether or not changes in hypertension status are *responding* to changes in lifestyle.

3. What the conditional proportion of sedentary volunteers who are not hypertensive?

(**A**) 40/60 **(B) 20/60** (**C**) 40/100 (**D**) 20/100

4. What are the odds of hypertension among volunteers who lead active lifestyles?

(**A**) 1/5 **(B) 1/7** (**C**) 5 **(D) 7**

Solution: Odds of "success" = (number of successes) / (number of failures)

5. The odds of developing hypertension while leading a sedentary lifestyle are 2. Which of the following provides the best interpretation of these odds?

(**A**) There is a 2% chance of developing hypertension while leading a sedentary lifestyle

(**B**) There is a 200% chance of developing hypertension while leading a sedentary lifestyle

(C) Among individuals with sedentary lifestyles: for every 1 person who is not hypertensive, there are 2 people who are hypertensive.

(**D**) Among individuals with sedentary lifestyles: for every 1 person who is hypertensive, there are 2 people who are not hypertensive.

Solution: Here's the heuristic I use:

$$\frac{(\text{Hypertensive})}{(\text{Not Hypertensive})} = 2 \implies (\text{Hypertensive}) = 2 \times (\text{Not Hypertensive})$$

So, indeed, for every one sedentary person who is not hypertensive there are 2 others (also with sedentary lifestyles) who are hypertensive.

6. Define the following two quantities:

- x = the odds ratio comparing the odds of hypertension among sedentary individuals to the the odds of hypertension among active individuals
- y = the odds ratio comparing the odds of hypertension among active individuals to the the odds of hypertension among sedentary individuals

Which of the following correctly captures the relationship between the quantities x and y?

(**A**) $x = y$ **(B)** $y = 1/x$

Solution: When we swap EITHER the notion of success OR the baseline, the odds ratio flips (i.e. becomes the reciprocal). If we swap BOTH the notion of success AND the baseline, the odds ratio returns to the original value.

There are two competing claims for why we observed the data we did:

- **Possibility 1:** activeness of lifestyle has no effect on hypertension.
- **Possibility 2:** leading active more lifestyles positively impacts cardiovascular health (i.e. leading a more active lifestyle is associated with a lower risk of hypertension)

To that end, define the following quantities:

- π_A, the population proportion of all US adults leading active lifestyles that are hypertensive
- π_S, the population proportion of all US adults leading sedentary lifestyles that are hypertensive

7. What are the correct null and alternative hypotheses for the study? (Pay close attention to the way the research question is phrased!)

(A) $\begin{bmatrix} \mathbf{H_0}: & \pi_A = \pi_S \\ \mathbf{H_A}: & \pi_A < \pi_S \end{bmatrix}$

(**B**) $\begin{bmatrix} H_0: & \pi_A = \pi_S \\ H_A: & \pi_A \neq \pi_S \end{bmatrix}$

(**C**) $\begin{bmatrix} H_0: & \pi_A = \pi_S \\ H_A: & \pi_A > \pi_S \end{bmatrix}$

Going forward, assume the null hypothesis is the "there is no effect" hypothesis and the alternative hypothesis is "leading a more active lifestyle decreases the chances of developing hypertension." Additionally, the theory-based p-value is around 0.000.

8. Would an approximate theory-based 95% confidence interval for $(\pi_S - \pi_A)$ contain zero or not, and why?

(**A**) The interval would contain zero, because zero is a plausible value for the difference in population proportions $(\pi_S - \pi_A)$.

(B) The interval would not contain zero, because the very small *p*-value of 0.000 indicates that we should reject the null hypothesis that activeness of lifestyle doesn't affect the risk of hypertension.

(**C**) The interval would contain zero, because the very small *p*-value of 0.000 indicates that we should fail to reject the null hypothesis that activeness of lifestyle doesn't affect the risk of hypertension.

Use the following setup for Questions 9 - 10

Shown below is a two-way table, with some entries missing.

	Control	**Treatment**	**Total**
Success	x	20	**30**
Failure	15	y	**30**
Total	**25**	**35**	**60**

Table 2: Two-Way Table of Results

9. What is the correct value of x?

(**A**) 5 **(B) 10** (**C**) 15 (**D**) 20 (**E**) 25

10. Let $\pi_C - \pi_C$ denote the difference in population proportions of successes in the Control and Treatment groups. Which of the following correctly provides a 95% confidence interval for $\pi_C - \pi_C$?

(**A**) $\left(\frac{y}{25} - 0.5714\right) \pm 2 \cdot \sqrt{\frac{\left(\frac{y}{25}\right)\left(1 - \frac{y}{25}\right)}{25} + \frac{(0.5714)(1 - 0.5714)}{35}}$

(B) $\left(\frac{x}{25} - 0.5714\right) \pm 2 \cdot \sqrt{\frac{\left(\frac{x}{25}\right)\left(1 - \frac{x}{25}\right)}{25} + \frac{(0.5714)(1 - 0.5714)}{35}}$

(**C**) $\left(\frac{y}{35} - 0.5714\right) \pm 2 \cdot \sqrt{\frac{\left(\frac{y}{25}\right)\left(1 - \frac{y}{25}\right)}{35} + \frac{(0.5714)(1 - 0.5714)}{25}}$

(**D**) $\left(\frac{x}{35} - 0.5714\right) \pm 2 \cdot \sqrt{\frac{\left(\frac{x}{35}\right)\left(1 - \frac{x}{35}\right)}{35} + \frac{(0.5714)(1 - 0.5714)}{25}}$